

Container Dashboard on FABRIC

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CSC 478 Cloud Engineering
Project Deliverable 1
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Chapter 1: Vision

In CSC 468 I built the Container Dashboard, a three-tier web app. It has an Nginx frontend, a Flask REST API, and a PostgreSQL database that stores a persistent log. It runs with Docker Compose on a single CloudLab virtual machine. That works, but everything depends on one host. If that VM goes down, the whole app goes down, and there's no way to spread the work across machines or scale up when needed.

For CSC 478, I want to keep the same application but move it onto a three-node Kubernetes cluster (RKE2) running on FABRIC. The whole environment, from the virtual machines to the running app, will be built by Jupyter notebooks. FABRIC slices are temporary and often need to be recreated, so automating every step makes the project reproducible. Kubernetes adds what a single Docker host can't: it restarts failed containers, moves work off a failed node, and scales parts of the app on demand.

Architecture

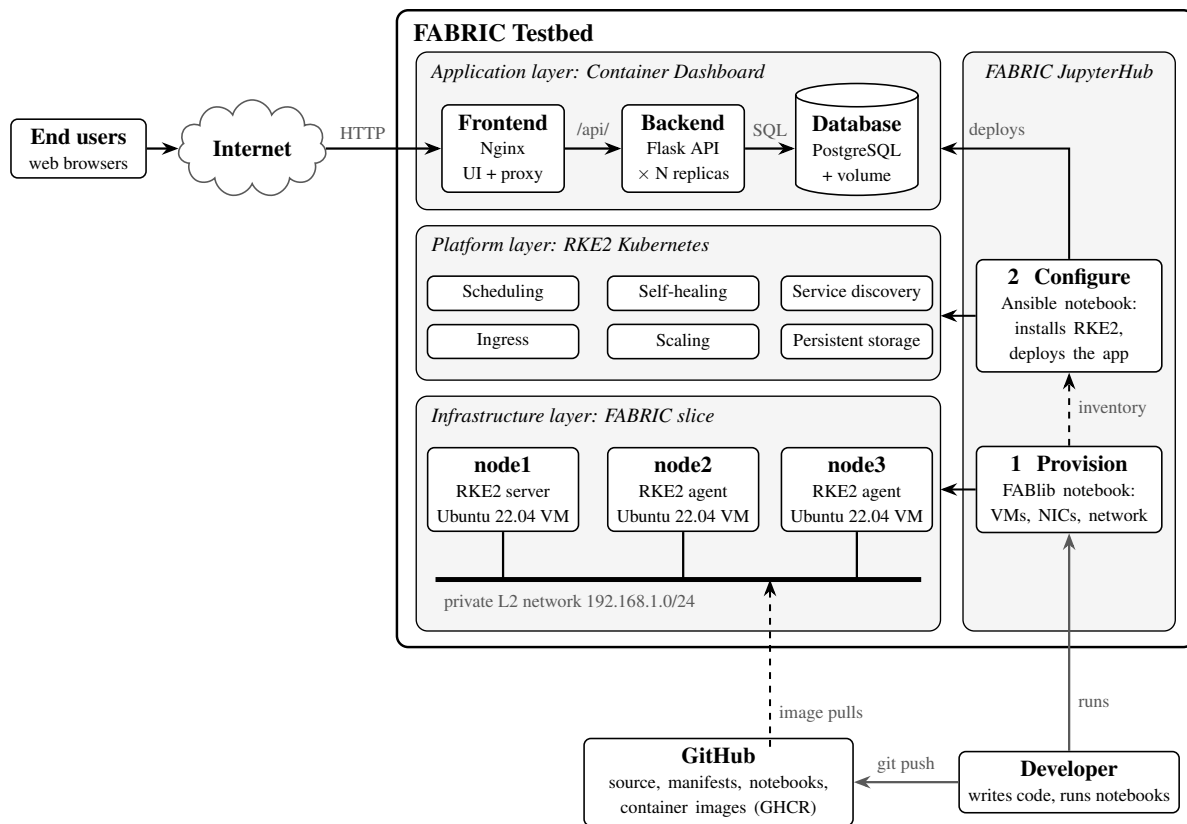


Figure 1. High-level architecture of the project.

The system is built in three layers:

- **Infrastructure layer:** three Ubuntu VMs on a private network, created by a FABlib notebook.
- **Platform layer:** an RKE2 Kubernetes cluster installed by an Ansible notebook. node1 is the server and node2 and node3 are agents.
- **Application layer:** the same frontend, backend, and database as before, now running as Kubernetes workloads.

Users reach the app through a single entry point, while the backend and database stay private inside the cluster. The code lives on GitHub, and the cluster pulls pre-built container images from there.

Chapter 2: Proposed Implementation

The project will be built in five steps:

1. **Container images:** Build the frontend and backend images and push them to GitHub's container registry so the cluster can pull them.
2. **Provisioning notebook:** Use FABlib to find a FABRIC site with enough resources and create three VMs connected by a private network.
3. **Cluster notebook:** Use Ansible to install RKE2 on node1, join node2 and node3 to the cluster, and add storage for the database.
4. **Deploy the app:** Apply Kubernetes files that run the database with persistent storage, run the backend and frontend with multiple copies, and expose only the frontend to users. The services keep the same names the app already uses (frontend, backend, db), so the code barely needs to change.
5. **Test it:** Confirm the app can be rebuilt from scratch, recovers when a container or node fails, scales up the backend, and keeps its data when the database restarts.

Ethan Charest

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Professional Summary

Computer Science undergraduate with hands-on experience in cloud infrastructure, automation, and system optimization. Focused on expanding technical expertise in cloud and DevOps technologies while applying problem-solving skills to build efficient, scalable solutions.

Education

West Chester University, West Chester, PA

Bachelor of Science in Computer Science | 2024 – Present

Delaware County Community College, Downingtown, PA

Computer Science Transfer Program | 2022 – 2024

Experience

Infrastructure and Cloud Services Intern

Wawa, Inc., Media, PA | 2024

- Deployed and configured virtual cloud servers in AWS to support enterprise infrastructure.
- Automated patching and connectivity processes across multiple vCenters using scripting tools, improving Jenkins pipeline efficiency.
- Performed server patching and updates to resolve critical vulnerabilities and enhance system performance.
- Allocated system resources and implemented feature updates based on operational requirements.
- Collaborated with cross-functional IT teams to optimize non-production cloud environments, reducing costs and environmental impact.

Customer Sales Associate

Wawa, Inc., Parkesburg, PA & West Chester, PA | 2020 – Present

- Delivered high-quality customer service in fast-paced retail and food service environments.

- Trained and mentored new employees, contributing to a collaborative team culture.
 - Maintained compliance with food safety and operational standards.
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Technical Skills

Programming Languages: Java, Python, JavaScript, Haskell, PowerShell, JSON, YAML

Cloud & DevOps: AWS, Azure, Docker, Jenkins, Ansible, Git, GitHub

Tools & Platforms: VMware, Citrix, Splunk, Datadog, Fabric

Frameworks & Systems: Jira, Confluence, ServiceNow, Workday